B-Tree Filesystem (BTRFS)

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Introduction

- BTRFS is a modern copy on write (CoW) filesystem for Linux aimed at implementing advanced features while also focusing on fault tolerance, repair and easy administration.
  - Snapshots which do not make a full copy of the files
  - Built-in volume management, support for software-based RAID 0, RAID 1 and others
  - Self-healing - checksums for data and metadata, automatic detection of silent data corruptions
  - Transparent data compression
  - Reflinks, fast and efficient file copies

```
# mount -o compress=zstd /dev/sdx /mnt

# cp --reflink=always source target
```

Why Btrfs?

- ext4 is the most common filesystem with basic functions.
- Btrfs has other capabilities, such as snapshots.
    - ZFS also has many capabilities while utilizing relatively high memory.

Subvolume

- A BTRFS subvolume is a part of filesystem with its own independent file/directory hierarchy and inode number **namespace**.
- Each Btrfs file system has a **top-level** subvolume with ID 5
  - The top-level subvolume has path `/` on the file system
  - Other subvolumes are nested below the top-level subvolume
  - By default, the top-level subvolume is mounted when mounting the file system
- Naming convention
  - name with “@” prefix
    - `@ for /`
    - `@www for /var/www`

Nested layout

- A subvolume is created inside another subvolume.
- You need to delete all children subvolumes before deleting the parent one.
- Subvolumes **don't** need to be mounted manually, and the children subvolumes will be mounted when the parent subvolume is mounted.
    - The mount options of parent subvolume will be applied to children subvolumes.

```
toplevel                    (volume root directory, not mounted)
\-- @                       (subvolume root directory, to be mounted at /)
    +-- @home               (subvolume root directory)
    +-- @var                (subvolume root directory)
        +-- @www            (subvolume root directory)
        +-- lib             (directory)
             \-- @postgresql (subvolume root directory)
```

Flat layout (1/2)

- All subvolumes are top level's children.
- All subvolumes need to be mounted manually.
  - Each of these subvolumes/mountpoints can be mounted with different options.

```
toplevel         (volume root directory, not to be mounted by default)
  +-- @          (subvolume root directory, to be mounted at /)
  +-- @home      (subvolume root directory, to be mounted at /home)
  +-- var        (directory)
  |   \-- @www   (subvolume root directory, to be mounted at /var/www)
  \-- @postgres  (subvolume root directory, to be mounted at /var/lib/postgresql)
```

Flat layout (2/2)

- `/etc/fstab` may look like below:

```
UUID=the-btrfs-fs-uuid     /                  btrfs subvol=@,defaults,autodefrag      0  0
UUID=the-btrfs-fs-uuid     /home              btrfs subvol=@home,defaults,noatime     0  0
UUID=the-btrfs-fs-uuid     /var/www           btrfs subvol=/var/@www,noatime           0  0
UUID=the-btrfs-fs-uuid     /var/lib/postgresql  btrfs subvol=/var/@postgres,noatime     0  0
```

Snapshot

- A snapshot is simply a subvolume that shares its data (and metadata) with some other subvolume, using Btrfs's **COW** capabilities.
  - Not recursive
  - `-r` option for read-only
  - There is no difference between the original subvolume and writable snapshot

```
btrfs subvolume snapshot [-r] <source subvolume> <destination subvolume>
```

LVM snapshot v.s. Btrfs snapshot

- Common
  - Copy on write
- LVM snapshot
  - Need to specify LV size when creating a snapshot
  - Write all snapshot volumes for same LV
    - Slow write speed and waste more space slightly
- Btrfs snapshot
  - No need to specify size for snapshot
  - Multiple snapshots (reflink) won’t waste performance & space

Snapshot rollback

- BTRFS has no built-in rollback command.
- The official documentation is unclear on the procedure for performing snapshot rollback.
  - Mount the snapshot subvolume and copy files to your destination folder
  - Snapshot the snapshot subvolume to another subvolume writable and remount
  - …

```
root@trixws:~# btrfs subvolume list -t /mnt
ID      gen     top level       path
--      ---     ---------       ----
261     47      5               home/@ray
267     39      5               snapshots/@homeray-1762502638

root@trixws:~# btrfs subvolume snapshot /mnt/snapshots/@homeray-1762502638 /mnt/home/@ray_new

root@trixws:~# mount -o subvol=/home/@ray_new /dev/sdb /home/ray
```

Example

- Migrate `home` directory from `ext4` to `btrfs`

Migrate home directory & snapshot

1. Install btrfs for your OS (`apt install btrfs-progs` on Debian)
2. Add a block device (partition/disk or LV, …) for btrfs
3. `mkfs.btrfs` for the new block device
4. Mount to temporary location and create btrfs subvolume for home
5. Copy home directory to the new subvolume
6. Mount new subvolume to `/home/USER`
7. Create snapshot subvolume for home subvolume

```
# Subvolume layout
toplevel             (volume root directory, not to be mounted by default)
  +-- home           (directory)
      \-- @USER      (subvolume root directory, to be mounted at /home/USER)
  +-- snapshots      (directory)
     \-- @homeUSER   (subvolume root directory, for storing /home/USER snapshot)
```

Migrate home directory (LV)

```
root@trixws:~# lvs
  LV       VG        Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  home-vol home-vg   -wi-a-----   3.99g
  root     trixws-vg -wi-ao---- <44.86g
  swap_1   trixws-vg -wi-ao----  <1.23g

root@trixws:~# lsblk
NAME                    MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                       8:0    0    4G  0 disk
├─sda1                    8:1    0    2G  0 part
│ └─home--vg-home--vol  254:2    0    4G  0 lvm
└─sda2                    8:2    0    2G  0 part
  └─home--vg-home--vol  254:2    0    4G  0 lvm
sdb                       8:16   0   48G  0 disk
├─sdb1                    8:17   0  976M  0 part /boot/efi
├─sdb2                    8:18   0  977M  0 part /boot
├─sdb3                    8:19   0   24G  0 part
│ ├─trixws--vg-root     254:0    0 44.9G  0 lvm  /
│ └─trixws--vg-swap_1   254:1    0  1.2G  0 lvm  [SWAP]
└─sdb4                    8:20   0 22.1G  0 part
  └─trixws--vg-root     254:0    0 44.9G  0 lvm  /
```

Migrate home directory (mkfs)

- Use `btrfs filesystem show` to show btrfs structure

```
root@trixws:~# mkfs.btrfs /dev/home-vg/home-vol
btrfs-progs v6.14
See https://btrfs.readthedocs.io for more information.

Performing full device TRIM /dev/home-vg/home-vol (3.99GiB) ...
…
Number of devices:  1
Devices:
   ID        SIZE  PATH
    1     3.99GiB  /dev/home-vg/home-vol

root@trixws:~# btrfs filesystem show
Label: none  uuid: 0ed693ae-28ac-44ce-8674-9059f685d8f1
        Total devices 1 FS bytes used 144.00KiB
        devid    1 size 3.99GiB used 536.00MiB path /dev/mapper/home--vg-home--vol
```

Migrate home directory (subvolume & copy)

```
root@trixws:~# mount /dev/home-vg/home-vol /mnt

root@trixws:~# mkdir -p /mnt/home

root@trixws:~# btrfs subvolume create /mnt/home/@ray
Create subvolume '/mnt/home/@ray'


root@trixws:~# cp -a /home/ray/. /mnt/home/@ray/
```

Migrate home directory (mount)

```
root@trixws:~# mount -o subvol=/home/@ray /dev/home-vg/home-vol /home/ray

root@trixws:~# df -h
Filesystem                      Size  Used Avail Use% Mounted on
/dev/mapper/trixws--vg-root      44G  1.2G   41G   3% /
...
/dev/mapper/home--vg-home--vol  4.0G  8.6M  3.5G   1% /mnt
/dev/mapper/home--vg-home--vol  4.0G  8.6M  3.5G   1% /home/ray

# ensure consistency after reboot
root@trixws:~# echo "UUID=0ed693ae-28ac-44ce-8674-9059f685d8f1 /home/ray btrfs subvol=/home/@ray 0 0" >> /etc/fstab
```

Migrate home directory (snapshot)

```
root@trixws:~# mkdir -p /mnt/snapshots

root@trixws:~# btrfs subvolume snapshot /mnt/home/@ray /mnt/snapshots/@homeray-$(date +%s)
Create subvolume '/mnt/snapshots/@homeray-1762502638'


root@trixws:~# btrfs subvolume list -t /mnt
ID      gen     top level       path
--      ---     ---------       ----
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267     39      5               snapshots/@homeray-1762502638
```

Reference

- Welcome to BTRFS documentation! — BTRFS documentation
- SysadminGuide - btrfs
- Btrfs - ArchWiki
- Btrfs migration - Debian Wiki
- Overview of file systems in Linux | Storage Administration Guide | SLES 15 SP7